

# HW6

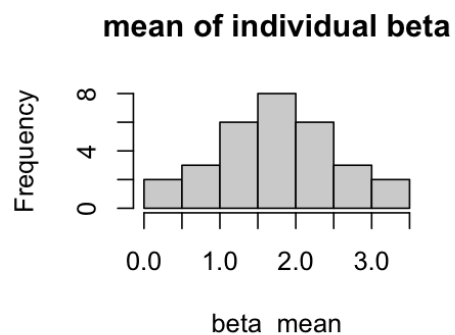
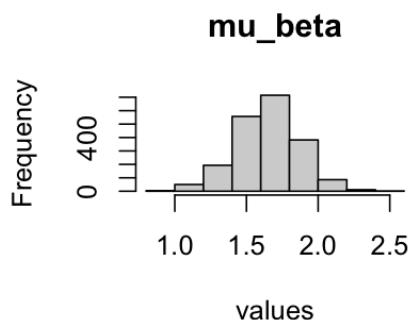
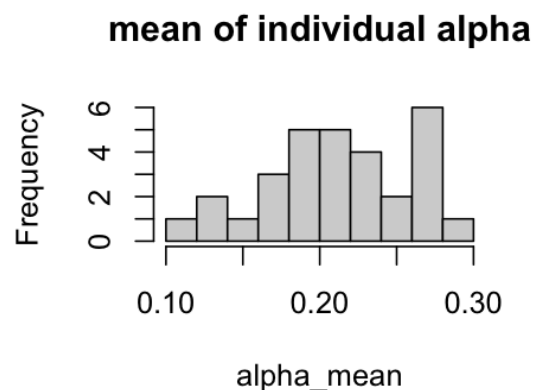
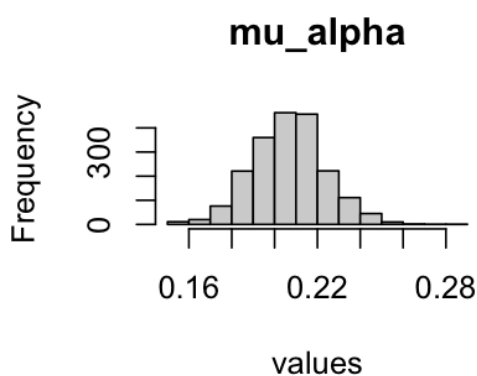
## Problem 1

### (0) Convergence check

All parameters appeared to converge successfully across the two chains. All  $\hat{R}$  values were smaller than 1.01, and no divergent transitions were observed.

### (2a) Posterior distributions of individual and group parameters

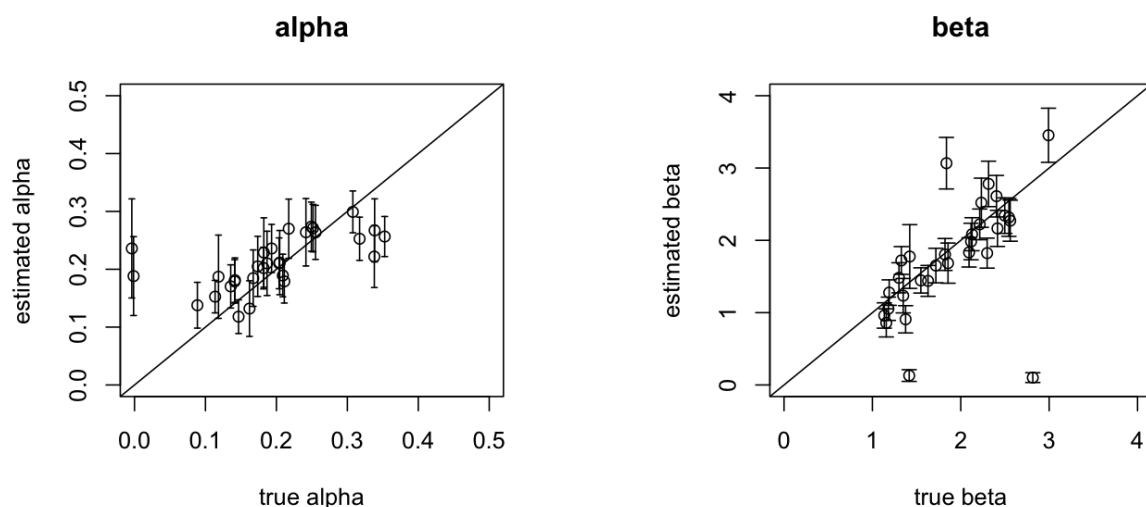
The figures on the left show the posterior distributions of the group parameters,  $\mu_\alpha$  and  $\mu_\beta$ , across chains. The figures on the right show the distributions of the posterior means of individual  $\alpha$  and  $\beta$ .



### (2.b). Comparison between true and estimated parameters

The figures below show the true individual parameters (x-axis) and estimated individual posterior means (y-axis) for  $\alpha$  (left) and  $\beta$  (right). For both  $\alpha$  and  $\beta$ , 18 out of 30 estimates included the true parameter value within posterior mean  $\pm 1$  SD.

The figures do not show a clear systematic bias toward overestimation or underestimation. In addition, the relative ordering of individuals was largely preserved. Based on these results, the parameters appear to be relatively well recovered.



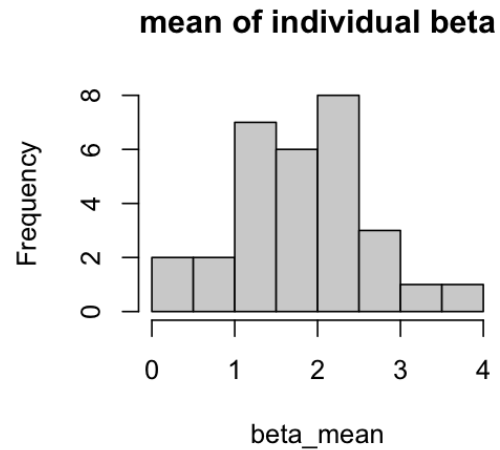
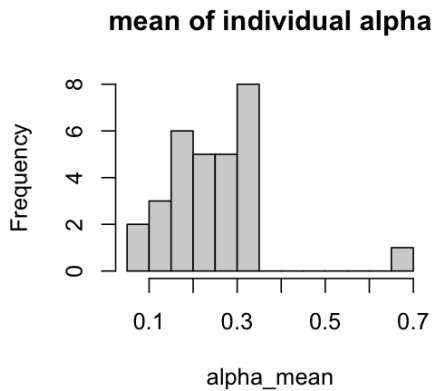
## Problem 2

### (0) Convergence check

One  $\beta$  parameter had an  $\hat{R}$  value greater than 1.01, and 5 divergent transitions were observed.

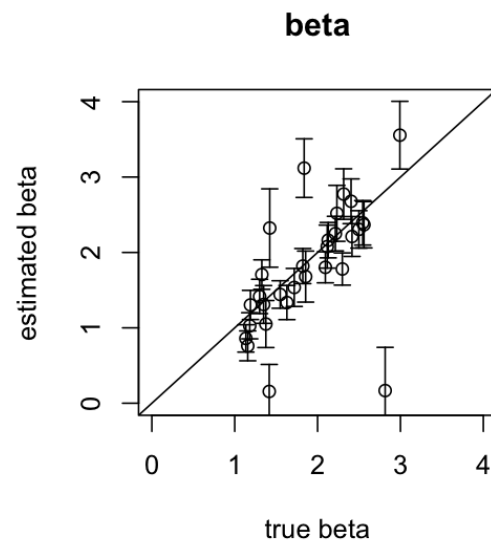
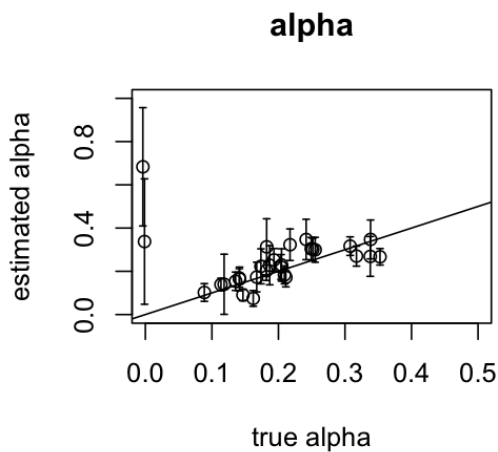
### (1) Posterior distributions of individual parameters (non-hierarchical model)

The figures below show the distributions of posterior means for individual  $\alpha$  (left) and  $\beta$  (right). Compared to Problem 1, the distributions are much more dispersed.



## (2) Comparison between true and estimated parameters

The figure below shows the true individual parameters (x-axis) and estimated individual parameters (y-axis) for alpha (left) and beta (right). 19 out of 30  $\alpha$  estimates and 17 out of 30  $\beta$  estimates included the true parameter value within posterior mean  $\pm 1$  SD.



## (3) Difference between Problem 1 and Problem 2

For the non-hierarchical model, some estimates deviated substantially from the true parameters, especially the two cases of strong  $\alpha$  overestimation. These extreme deviations were reduced in the hierarchical model.

As a result, the mean squared deviation of  $\alpha$  estimates was much lower in the hierarchical model (individual: 0.02 vs. hierarchical: 0.005). The mean squared deviation of  $\beta$  estimates was also slightly lower in the hierarchical model (individual: 0.43 vs. hierarchical: 0.41).

This is likely because the hierarchical model assumes that individual parameters are drawn from a common group-level distribution. In this simulation, this assumption matches the true data-generating process ( $\alpha_i \sim N(0.2, 0.08)$ ,  $\beta_i \sim N(2, 0.7)$ ). Therefore, shrinkage in the hierarchical model helped reduce extreme deviations from the true parameters

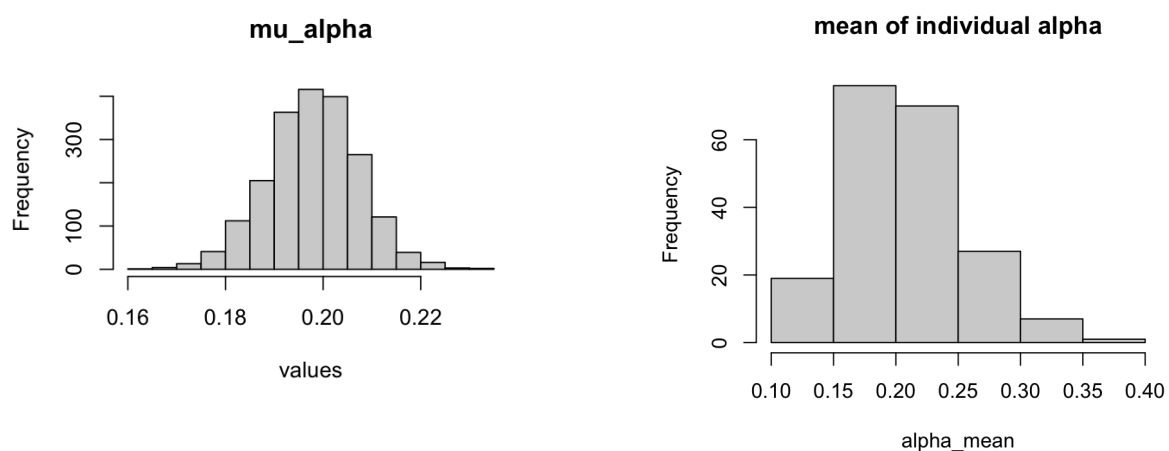
### Problem 3

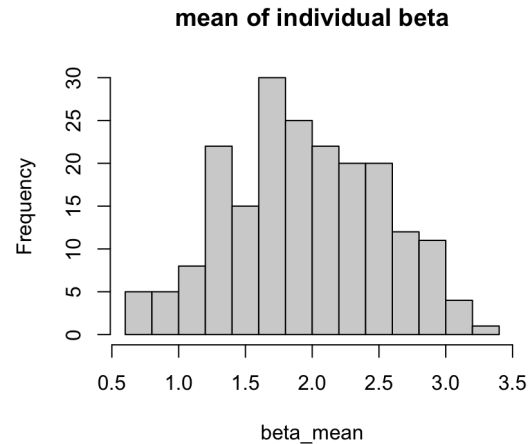
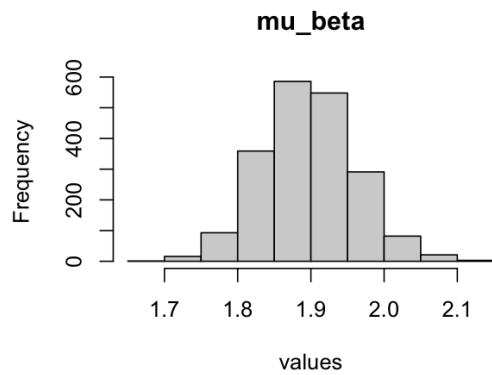
#### (0) Convergence check

All parameters appeared to converge successfully across the two chains. All  $\hat{R}$  values were smaller than 1.01, and no divergent transitions were observed.

#### (2) Posterior distributions of individual and group parameters

The figures on the left show the distribution of group parameters,  $\mu_\alpha$  and  $\mu_\beta$  across chains. The figures on the right show the distribution of posterior means for individual  $\alpha$  and  $\beta$ .

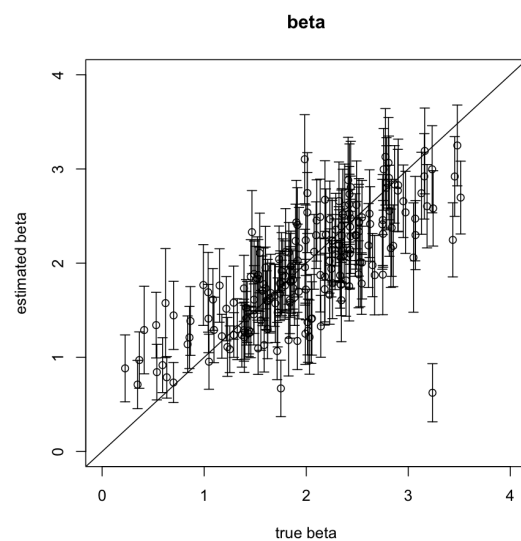
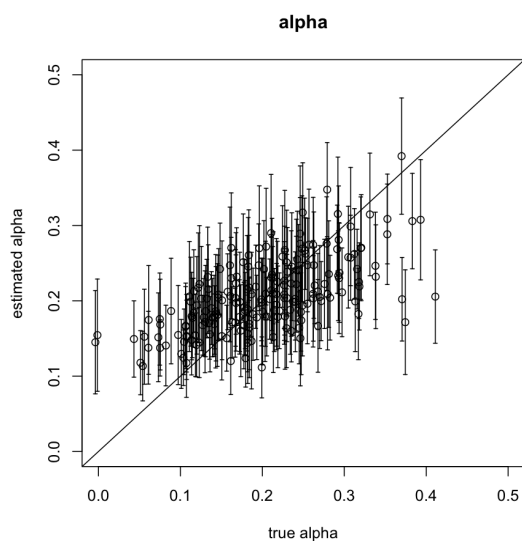




### (3) Comparison between true and estimated parameters

The figures below show the true individual parameters (x-axis) and estimated posterior means (y-axis) for  $\alpha$  (left) and  $\beta$  (right). Among 200 subjects, 129  $\alpha$  estimates and 131  $\beta$  estimates included the true parameter value within posterior mean  $\pm 1$  SD.

The figures do not show a clear systematic bias toward overestimation or underestimation.



### (4) The difference in shrinkage between $\alpha$ and $\beta$

Stronger shrinkage was observed for  $\alpha$  than for  $\beta$ . In addition, compared to Problem 1, the distribution of posterior means for individual  $\alpha$  became

noticeably narrower. (This is clear when comparing “mean of individual  $\alpha$  histogram”.)

This may be because learning rates mainly influence behavior when sufficiently large prediction errors occur. However, after the early learning phase, simulated subjects are likely to repeatedly choose the better option with reward probability 0.8. Therefore, large prediction errors occur relatively rarely, mainly when the rare loss outcome occurs or when the subject explores the lower-value option.

As a result, the individual variability in learning rates does not strongly affect the likelihood, so the likelihood contains relatively limited information about individual learning rates. This increases the influence of the group-level distribution. Since the number of trials decreased from 300 to 100 compared to Problem 1, these informative prediction errors became even more sparse.

In contrast,  $\beta$  consistently affects choice behavior throughout the task, regardless of prediction error size. Therefore, even though shrinkage increased overall, the degree of shrinkage for  $\beta$  was smaller than that for  $\alpha$ .

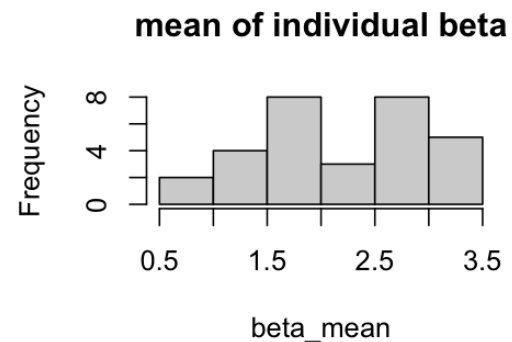
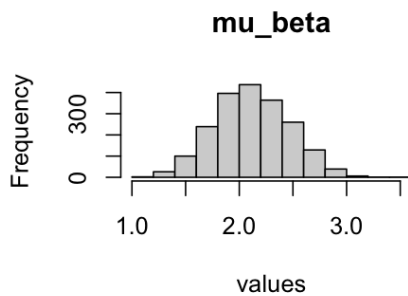
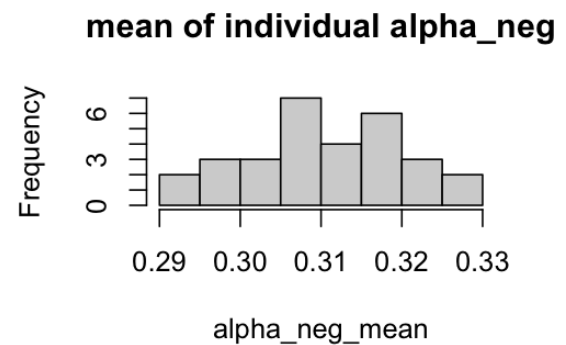
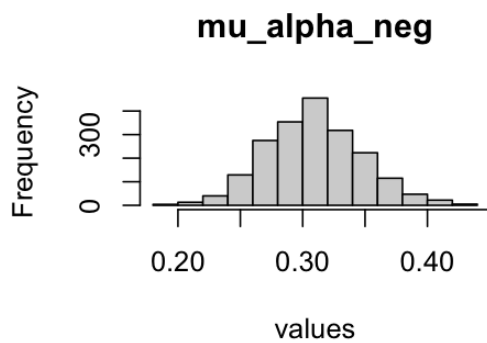
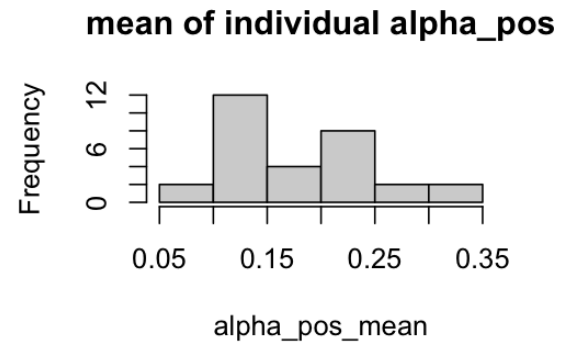
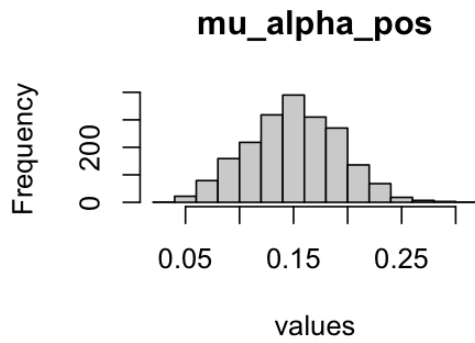
## Problem 4

### (0) Convergence check

The model appeared to converge reasonably well. No divergent transitions were observed, and all  $\hat{R}$  values were smaller than 1.01 except for two individual parameters (  $\alpha[1]$  and  $\beta[15]$ ), whose values were only slightly greater than 1.01.

### (1) Posterior distribution of individual and group parameters

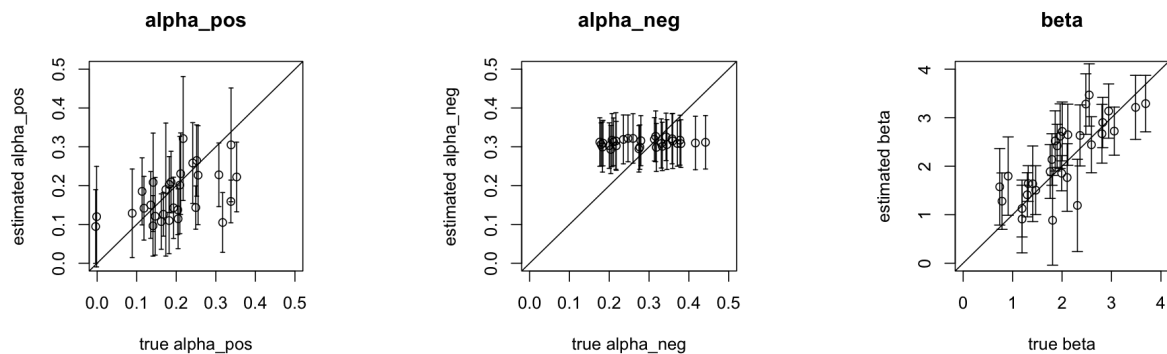
The figures on the left show the posterior distributions of group parameters,  $\mu_{\alpha_{pos}}$ ,  $\mu_{\alpha_{neg}}$ ,  $\mu_{\beta}$  across chains. The figures on the right show the distributions of posterior means of individual  $\alpha_{pos}$ ,  $\alpha_{neg}$  and  $\beta$



## (2) Comparison of true and estimated parameters

The figures below show the the true individual parameters (x-axis) and estimated posterior means (y-axis) for  $\alpha_{pos}$ ,  $\alpha_{neg}$  and  $\beta$ . While 23  $\alpha_{pos}$  estimates and 23  $\beta$  estimates included the true parameter within posterior mean  $\pm 1$  SD, only 14  $\alpha_{neg}$  estimates did. In addition,  $\alpha_{neg}$  estimates did not preserve the relative order of true individual parameters, and the distribution of posterior means across individuals was much narrower than the distribution of

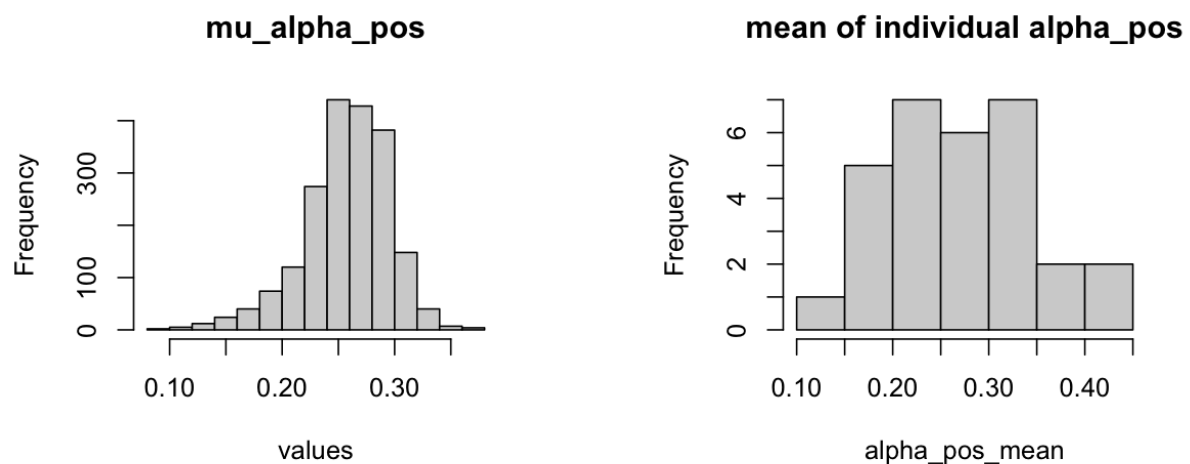
true parameters. Based on these facts,  $\alpha_{pos}$  and  $\beta$  appear to be relatively well recovered, whereas  $\alpha_{neg}$  doesn't

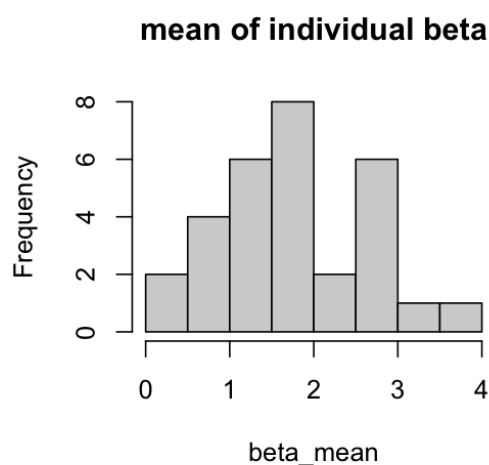
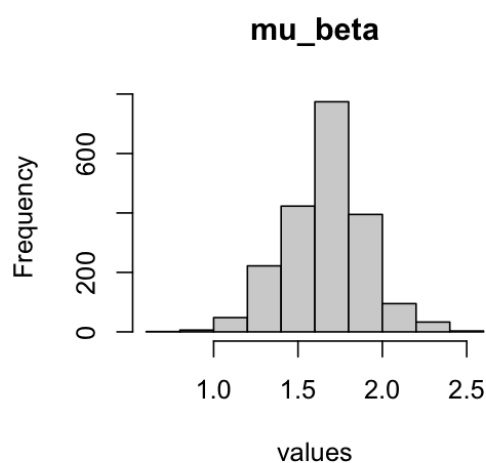
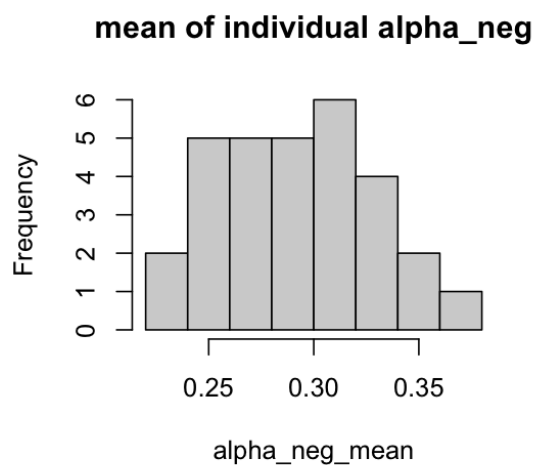
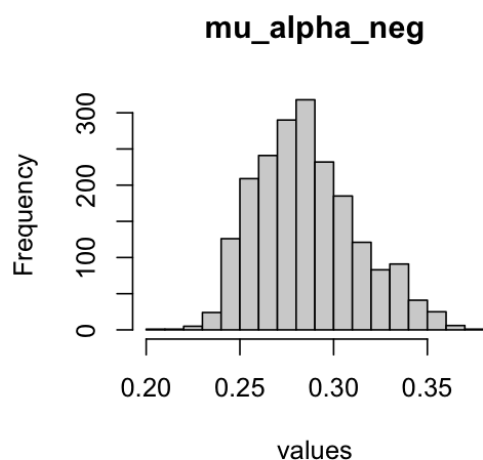


### (3) Posterior distributions and comparison between true and estimated parameters

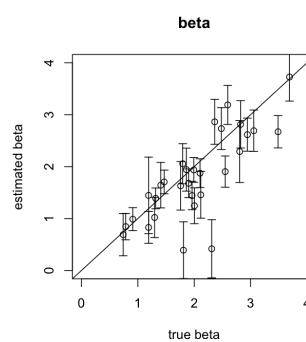
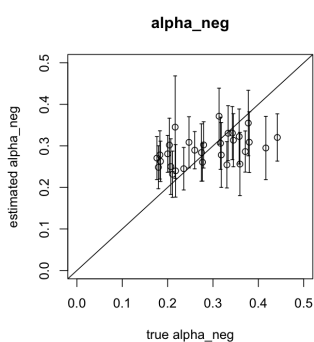
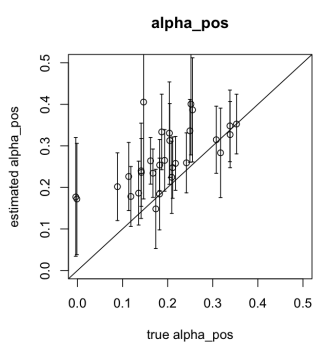
First, the model did not converge well. 69 parameters had  $\hat{R}$  values greater than 1.01, and 71 divergent transitions were observed.

The figures on the left show the posterior distributions of the group parameters,  $\mu_{\alpha_{pos}}$ ,  $\mu_{\alpha_{neg}}$ ,  $\mu_{\beta}$  across chains. The figures on the right show the distributions of posterior means for individual  $\alpha_{pos}$ ,  $\alpha_{neg}$  and  $\beta$ .





The figures below show the the true individual parameters (x-axis) and estimated posterior means (y-axis) for  $\alpha_{pos}$ ,  $\alpha_{neg}$  and  $\beta$ . 15  $\alpha_{pos}$  estimates, 15  $\alpha_{neg}$  estimates, and 18  $\beta$  estimates included the true parameter value within posterior mean  $\pm 1$  SD.



#### (4) Difference between (2) and (3)

The distribution of  $\alpha_{neg}$  became more dispersed in (3) compared to (2).

There were two major changes in the simulation settings:

1. the number of trials increased.
2. the reward probabilities became more uncertain (0.3 vs. 0.6 instead of 0.2 vs. 0.8).

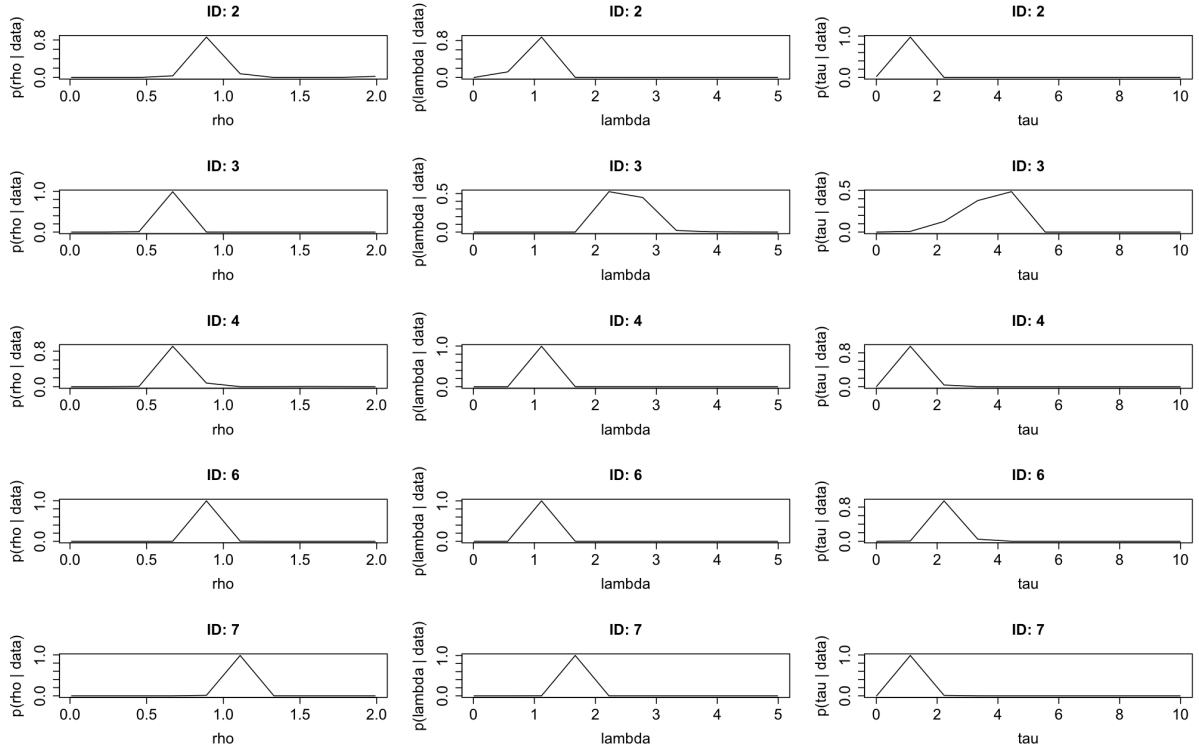
Because of this, negative prediction errors occurred more frequently throughout the task. As a result, the influence of individual variability on the likelihood becomes greater, so the likelihood contained more information about individual differences in  $\alpha_{neg}$ . This reduces the influence of the group-level distribution, leading to weaker shrinkage of  $\alpha_{neg}$  in (3).

Another noticeable difference was the overestimation of  $\alpha_{pos}$ . This may be related to the increased uncertainty of the reward structure in (3). Since the reward probabilities were closer to each other, the optimal option became less consistently rewarding, and positive prediction errors continued to occur throughout the task. Under this setting, rapid updating after positive outcomes may have been overemphasized during estimation, leading to larger estimates of  $\alpha_{pos}$ .

## Problem 5

(1) posterior distribution using 10 grids per parameter

The figure below shows the distributions of individual parameters derived from, using 10 grids per parameter and uniform priors ( $\rho \sim \text{uniform}(0, 2)$ ,  $\lambda \sim \text{uniform}(0, 5)$ ,  $\tau \sim \text{uniform}(0, 10)$ ).



## (2) posterior distribution using 10 grids per parameter

The figure below shows the distributions of individual parameters derived from, using 30 grids per parameter and the same uniform priors.

